

ECO 529: Macro-Finance: Tools, Models, and Ideas

Lecture 5: Recursive Optimal Control

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“Finding the State is an Art.”

Tom Sargent

Goals

- ▶ Study optimal control of continuous time stochastic systems
- ▶ Multiple approaches:
 1. **Dynamic programming** (this course)
(Easiest to solve but requires problem to be recursive)
 2. Pontryagin's maximum principle
(Required when problem cannot be expressed recursively)
 3. Martingale Approach
(Useful when obvious SDF for pricing)
- ▶ Derive Hamilton-Jacobi-Bellman equation (HJBE)
- ▶ Applications: Consumption & saving, portfolio choice

References

- ▶ Focus on economic examples: [Björk, 2012]
- ▶ Relatively user friendly technical introductions:
[Pham, 2009], [Chang, 2004] and [Hanson, 2007]
- ▶ Technical references:
[Fleming and Soner, 2006], [Fleming and Rishel, 2012] or [Yong and Zhou, 1999]
- ▶ Additional references on viscosity solutions:
Lions (1983), Crandall (1997), Bressan (2011), Yu (2011), Lions et al. (1992)

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Application: Portfolio Problem

Application: The Consumption Saving Problem

- ...with Poisson Shocks

- ...with a Borrowing Constraint

Numerical Techniques

- Finite Difference

- Deep Learning

Agent Control Problem

Agent's **stochastic optimal control problem** is to choose control, c , to solve:

$$V_0(x) \equiv \sup_{c \in \mathcal{C}(x)} \mathbb{E} \left[\int_0^\infty e^{-\rho s} u(c_s) ds \right]$$

subject to the dynamic constraint on the evolution of their **state variable** X_t :

$$dX_t = \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t, \quad t \in (0, \infty), \quad X_0 = x \quad (1)$$

- ▶ $\rho > 0$: is agent's discount function,
- ▶ c : is the agent's **control process**; if a control process can be written in the form $c_s = c(X_s)$ for some function, c , then it is called a **Markov control**,
- ▶ u : is agent flow utility function,
- ▶ μ_t : is the drift of the state variable,
- ▶ σ_t : is the volatility of the state variable, and
- ▶ W_t : is a univariate Brownian motion process.

Hamilton-Jacobi-Bellman Equation

Proposition (Necessity of the HJBE)

- Suppose that $X_t = x$. Define the (current) value function at time t by:

$$V_t(x) := \sup_{c \in \mathcal{C}(x)} \mathbb{E}_{t,x} \left[\int_t^\infty e^{-\rho(s-t)} u(c_s) ds \right]$$

where $\mathbb{E}_{t,x}[\cdot] := \mathbb{E}[\cdot | X_t = x]$ and X_t is constrained to follow (1).

- Suppose that an optimal control $\hat{c}$ exists. Suppose that $V_t(x)$ exists (is finite) and is $C^{1,2}$ (continuously once differentiable in t and twice continuously differentiable in x).
- Then, $V_t(x)$ satisfies the Hamilton-Jacobi-Bellman Equation (HJBE):

$$\rho V_t(x) = \max_{c \in \mathcal{C}(x)} \left\{ u(c) + \mu_t(x, c) \partial_x V_t(x) + \frac{1}{2} \sigma_t^2(x, c) \partial_{xx} V_t(x) \right\} + \partial_t V_t(x)$$

Steps For Deriving the HJBE

0. Define notation and state assumptions
1. Apply the Dynamic Programming Principle
2. Apply Itô's Lemma
3. Take Expectations
4. Divide by h and take the limit as $h \rightarrow 0$
5. Impose Optimality

0. Define Value functions

- Recall definition of the (current) value function:

$$V_t(x) \equiv \sup_{c \in \mathcal{C}} \mathbb{E}_{t,x} \left[\int_t^\infty e^{-\rho(s-t)} u(c_s) ds \right]$$

- Define the present discounted value function:
(which will end up being easier to work with)

$$V_t^P(x) \equiv e^{-\rho t} V_t(x) = \sup_{c \in \mathcal{C}} \mathbb{E}_{t,x} \left[\int_t^\infty e^{-\rho s} u(c_s) ds \right]$$

1. Apply Dynamic Programming Principle

- Let c be an arbitrary control and suppose agent chooses:

$$c_s^* = \begin{cases} c_s, & s \in [t, t+h], & \text{arbitrary control} \\ \hat{c}_s, & s \in (t+h, \infty], & \text{optimal control} \end{cases}$$

- Then, by the law of conditional iterated expectations, we have:

$$\begin{aligned} V_t^P(x) &\geq \mathbb{E}_{t,x} \left[\int_t^\infty e^{-\rho s} u(c_s^*) ds \right] \\ &= \mathbb{E}_{t,x} \left[\int_t^{t+h} e^{-\rho s} u(c_s) ds + \mathbb{E}_{t+h, X_{t+h}} \left[\int_{t+h}^\infty e^{-\rho s} u(\hat{c}_s) ds \right] \right] \\ &= \mathbb{E}_{t,x} \left[\int_t^{t+h} e^{-\rho s} u(c_s) ds + V_{t+h}^P(X_{t+h}) \right] \end{aligned}$$

Rearranging gives:

$$0 \geq \mathbb{E}_{t,x} \left[\int_t^{t+h} e^{-\rho s} u(c_s) ds \right] + \mathbb{E}_{t,x} [V_{t+h}^P(X_{t+h}) - V_t^P(x)]$$

2. Apply Ito's Lemma

- By Ito's Lemma we have:

$$\begin{aligned}dV_t^P(X_t) &= \partial_t V_t^P(X_t)dt + \mu_t(X_t, c_t)\partial_x V_t^P(x)dt + \frac{1}{2}\sigma_t^2(X_t, c_t)\partial_{xx} V_t^P(X_t)dt \\ &\quad + \sigma_t(X_t, c_t)\partial_x V_t^P(X_t)dW_t\end{aligned}$$

- So, the difference is:

$$\begin{aligned}&V_{t+h}^P(X_{t+h}) - V_t^P(x) \\ &= \int_t^{t+h} \left(\partial_s V_s^P(X_s) + \mu_s(X_s, c_s)\partial_x V_s^P(X_s) + \frac{1}{2}\sigma_s^2(X_s, c_s)\partial_{xx} V_s^P(X_s) \right) ds \\ &\quad + \int_t^{t+h} \sigma_s(X_s, c_s)\partial_x V_s^P(X_s)dW_s\end{aligned}$$

3. Take Expectations

$$\begin{aligned} & \mathbb{E}_{t,x} [V_{t+h}^P(X_{t+h}) - V_t^P(x)] \\ &= \mathbb{E}_{t,x} \left[\int_t^{t+h} \left(\partial_s V_s^P(X_s) + \mu_s(X_s, c_s) \partial_x V_s^P(X_s) + \frac{1}{2} \sigma_s^2(X_s, c_s) \partial_{xx} V_s^P(X_s) \right) ds \right] \\ & \quad + \mathbb{E}_{t,x} \left[\int_t^{t+h} \sigma_s(X_s, c_s) \partial_x V_s(X_s) dW_s \right] \\ &= \mathbb{E}_{t,x} \left[\int_t^{t+h} \left(\partial_s V_s^P(X_s) + \mu_s(X_s, c_s) \partial_x V_s^P(X_s) + \frac{1}{2} \sigma_s^2(X_s, c_s) \partial_{xx} V_s^P(X_s) \right) ds \right] \end{aligned}$$

4. Divide by h and take the limit as $h \rightarrow 0$

- Substitute the expression from Step 3 into the equation from Step 1:

$$\begin{aligned} 0 &\geq \mathbb{E}_{t,x} \left[\int_t^{t+h} e^{-\rho s} u(c_s) ds \right] + \mathbb{E}_{t,x} [V_{t+h}^P(X_{t+h}) - V_t^P(x)] \\ &= \mathbb{E}_{t,x} \left[\int_t^{t+h} e^{-\rho s} u(c_s) ds + \int_t^{t+h} \left(\partial_s V_s^P(X_s) + \mu_s(X_s, c_s) \partial_x V_s^P(X_s) \right. \right. \\ &\quad \left. \left. + \frac{1}{2} \sigma_s^2(X_s, c_s) \partial_{xx} V_s^P(X_s) \right) ds \right] \end{aligned}$$

- Divide by h to get:

$$\begin{aligned} 0 &\geq \mathbb{E}_{t,x} \left[\frac{1}{h} \int_t^{t+h} e^{-\rho s} u(c_s) ds + \frac{1}{h} \int_t^{t+h} \left(\partial_s V_s^P(X_s) + \mu_s(X_s, c_s) \partial_x V_s^P(X_s) \right. \right. \\ &\quad \left. \left. + \frac{1}{2} \sigma_s^2(X_s, c_s) \partial_{xx} V_s^P(X_s) \right) ds \right] \end{aligned}$$

- Taking the limit as $h \rightarrow 0$ gives:

$$0 \geq e^{-\rho t} u(c) + \partial_t V_t^P(x) + \mu_t(x, c) \partial_x V_t^P(x) + \frac{1}{2} \sigma_t^2(x, c) \partial_{xx} V_t^P(x)$$

5. Convert to $V_t(x)$ and Impose Optimality

- We have that:

$$V_t^P(x) = e^{-\rho t} V_t(x) \quad \Rightarrow \quad \partial_t V_t^P(x) = -\rho e^{-\rho t} V_t(x) + e^{-\rho t} \partial_t V_t(x)$$

- Substituting this into the HJBE gives:

$$0 \geq e^{-\rho t} u(c) - \rho e^{-\rho t} V_t(x) + e^{-\rho t} \partial_t V_t(x) + \mu_t(x, c) e^{-\rho t} \partial_x V(x) + \frac{1}{2} \sigma^2(x, c) e^{-\rho t} \partial_{xx} V(x)$$

which can be arranged as:

$$\rho V_t(x) \geq u(c) + \mu_t(x, c) \partial_x V_t(x) + \frac{1}{2} \sigma_t^2(x, c) \partial_{xx} V_t(x) + \partial_t V_t(x)$$

- The inequality was for arbitrary c . It binds when $c = \hat{c}$. Thus:

$$\rho V_t(x) = \max_{c \in \mathcal{C}(x)} \left\{ u(x, c) + \mu_t(x, c) \partial_x V_t(x) + \frac{1}{2} \sigma_t^2(x, c) \partial_{xx} V_t(x) \right\} + \partial_t V_t(x). \quad \square$$

Extra: Taking the Limit as $h \rightarrow 0$ (step 4)

- We have that:

$$\Rightarrow 0 \geq \frac{1}{h} \mathbb{E}_{t,x} \left[\int_t^{t+h} e^{-\rho s} u(c_s) ds \right] + \frac{1}{h} \mathbb{E}_{t,x} [V_{t+h}^P(X_{t+h}) - V_t^P(x)]$$

- Consider the first term. Define:

$$G(t) \equiv \int_0^t e^{-\rho s} u(c_s) ds$$

Then, taking the limit as $h \rightarrow 0$ gives (by the FTC):

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{1}{h} \mathbb{E}_{t,x} \left[\int_t^{t+h} e^{-\rho s} u(c_s) ds \right] &= \mathbb{E}_{t,x} \left[\lim_{h \rightarrow 0} \frac{G(t+h) - G(t)}{h} \right] \\ &= \mathbb{E}_{t,x} \left[\frac{\partial}{\partial t} G(t) \right] = \mathbb{E}_{t,x} \left[\frac{\partial}{\partial t} \int_0^t e^{-\rho s} u(c_s) ds \right] = e^{-\rho t} u(c_t) \end{aligned}$$

- Consider the second term. Can apply the same approach.

Comparison to Discrete Time

- Discrete time:

$$V_t(x) = \max_{c_t} \{u(c_t) + \beta \mathbb{E}_{t,x} [V_{t+1}(X_{t+h})]\}$$

- Continuous time:

$$\begin{aligned} V_t^P(x) &= \max_c \left\{ \mathbb{E}_{t,x} \left[\int_t^{t+h} e^{-\rho s} u(c_s) ds + V_{t+h}^P(X_{t+h}) \right] \right\} \\ \Rightarrow 0 &= \max_c \left\{ \mathbb{E}_{t,x} \left[\int_t^{t+h} e^{-\rho s} u(c_s) ds \right] + \mathbb{E}_{t,x} [V_{t+h}^P(X_{t+h}) - V_t^P(x)] \right\} \end{aligned}$$

after taking the limit as $h \rightarrow 0$, the expectation can be evaluated

$$\Rightarrow 0 = \max_c \left\{ e^{-\rho t} u(c) + \partial_t V_t^P(x) + \mu_t(x, c) \partial_x V_t^P(x) + \frac{1}{2} \sigma_t^2(x, c) \partial_{xx} V_t^P(x) \right\}$$

Continuous time Brownian motion replaces expectations by derivatives.

Hamilton-Jacobi-Bellman Equation

Corollary (Necessity of the HJBE: Stationary Case (where $\partial_t V_t(x) = 0$))

- Suppose that X follows a diffusion with drift and volatility functions that are independent of time:

$$dX_t = \mu(X_t, c_t)dt + \sigma(X_t, c_t)dW_t$$

- Suppose that $X_t = x$. Define the (current) value function at time t by:

$$V(x) := \sup_{c \in \mathcal{C}(x)} \mathbb{E}_{t,x} \left[\int_t^\infty e^{-\rho(s-t)} u(c_s) ds \right]$$

where $\mathbb{E}_{t,x}[\cdot] := \mathbb{E}[\cdot | X_t = x]$. Suppose $V(x)$ exists and is $C^{1,2}$. Then, $V(x)$ satisfies the Hamilton-Jacobi-Bellman Equation (HJBE):

$$\rho V(x) = \max_{c \in \mathcal{C}(x)} \left\{ u(c) + \mu(x, c) \partial_x V(x) + \frac{1}{2} \sigma^2(x, c) \partial_{xx} V(x) \right\}$$

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Application: Portfolio Problem

Application: The Consumption Saving Problem

- ...with Poisson Shocks

- ...with a Borrowing Constraint

Numerical Techniques

- Finite Difference

- Deep Learning

Univariate Poisson Processes

- ▶ Now, suppose that $X_t \in \mathbb{R}$ and follows the Itô process:

$$dX_t = \mu_t(X_t, c_t)dt + h_t(X_t, c_t)dN_t$$

where

- ▶ N_t is an 1-dimensional Poisson motion with rate $\lambda(x)$,
- ▶ $\mu_t : \mathbb{R} \times C \rightarrow \mathbb{R}$ is the drift, and
- ▶ $h_t : \mathbb{R} \times C \rightarrow \mathbb{R}$ is the jump size.

HJBE: Poisson Shocks; Infinite Horizon

Proposition (Necessity of the HJBE)

Define the (current) value function at time t by:

$$V_t(x) := \sup_{c \in \mathcal{C}(t,x)} \mathbb{E}_{t,x} \left[\int_t^\infty e^{-\rho(s-t)} u(c_s) ds \right].$$

Suppose $V_t(x)$ exists and is $C^{1,2}$. Then $V_t(x)$ satisfies the HJBE:

$$\rho V_t(x) = \max_{c \in \mathcal{C}(x)} \left\{ u(c) + \mu_t(x, c) \partial_x V_t(x) + \lambda (V_t(x + h_t(x)) - V_t(x)) \right\} + \partial_t V_t(x)$$

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Finite Horizon

- ▶ Same set-up as before, except for the following changes.
- ▶ Finite time horizon: $t \in [0, T]$.
- ▶ Let $U : \mathbb{R} \rightarrow \mathbb{R}$ be measurable termination utility function that is (i) lower bounded and (ii) satisfies the quadratic growth condition $|U(x)| \leq K(1 + |x|^2)$, $\forall x \in \mathbb{R}$ for some $K > 0$.
- ▶ For initial condition $X_0 = x$, the **value function** is defined by:

$$V(x) := \sup_{c \in \mathcal{C}(x)} \mathbb{E} \left[\int_0^T e^{-\rho s} u(c_s) ds + U(X_T^x) \right] \quad (2)$$

where $\mathcal{C}(x)$ denotes the subset of controls such that:

$$\mathbb{E} \left[\int_0^T e^{-\rho s} |u(c_s)| ds \right] < \infty$$

Summary: Control Problem

The agent's **finite horizon stochastic optimal control problem** is to choose control, c , to solve:

$$V(x) := \sup_{c \in \mathcal{C}(x)} \mathbb{E} \left[\int_0^T e^{-\rho s} u(c_s) ds + U(X_T^x) \right]$$

subject to the dynamic constraint on the evolution of X_t :

$$\begin{aligned} dX_t &= \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t, \quad t \in (0, T) \\ X_0 &= x \end{aligned}$$

Hamilton-Jacobi-Bellman Equation

Proposition (Necessity of the HJBE)

- Suppose that X follows a diffusion:

$$dX_t = \mu_t(X_t, c_t)dt + \sigma_t(X_t, c_t)dW_t \quad (3)$$

- Suppose that $X_t = x$. Define the (current) value function at time t by:

$$V_t(x) := \sup_{c \in \mathcal{C}(x)} \mathbb{E}_{t,x} \left[\int_t^T e^{-\rho(s-t)} u(c_s) ds + U(X_T^x) \right]$$

where $\mathbb{E}_{t,x}[\cdot] := \mathbb{E}[\cdot | X_t = x]$ and X_t is constrained to follow (3). Suppose $V_t(x)$ exists and is $C^{1,2}$. Then, $V_t(x)$ satisfies the Hamilton-Jacobi-Bellman Equation (HJBE):

$$\begin{aligned} \rho V_t(x) &= \max_{c \in \mathcal{C}(x)} \left\{ u(c) + \mu_t(x, c) \partial_x V_t(x) + \frac{1}{2} \sigma_t^2(x, c) \partial_{xx} V_t(x) \right\} + \partial_t V_t(x) \\ V_T(x) &= U(x) \end{aligned}$$

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Application: Portfolio Problem

Application: The Consumption Saving Problem

...with Poisson Shocks

...with a Borrowing Constraint

Numerical Techniques

Finite Difference

Deep Learning

Higher Dimensional Processes

- ▶ Now, suppose that $X_t \in \mathbb{R}^n$ and follows the Itô process:

$$dX_t = \mu_t(X_t)dt + \sigma_t(X_t)dW_t$$

where

- ▶ W_t is an m dimensional Brownian motion,
 - ▶ $\mu_t(x)$ is n dimensional, and
 - ▶ $\sigma_t(x)$ is $n \times m$ dimensional.
-
- ▶ Define:
 - ▶ $\Sigma_t(x) := \sigma_t(x)\sigma_t(x)^T$
 - ▶ $\Sigma_{t,jk}$ to refer to the j, k element of Σ_t

HJBE: Infinite Horizon

Proposition (Necessity of the HJBE)

Define the (current) value function at time t by:

$$V_t(x) := \sup_{c \in \mathcal{C}(t,x)} \mathbb{E}_{t,x} \left[\int_t^\infty e^{-\rho(s-t)} u(c_s) ds \right].$$

Suppose $V_t(x)$ exists and is $C^{1,2}$. Then $V_t(x)$ satisfies the HJBE:

$$\begin{aligned} \rho V_t(x) &= \max_{c \in \mathcal{C}(x)} \left\{ u(c) + \sum_{j=1}^n \mu_{t,j}(x, c) \partial_{x_j} V_t(x) + \frac{1}{2} \sum_{i,j=1}^n \Sigma_{t,jk}^2(x, c) \partial_{x_j x_k} V_t(x) \right\} + \partial_t V_t(x) \\ &= \sup_{c \in \mathcal{C}(x)} \left\{ u(c) + \mu_t(x, c) \cdot D_x V_t(x) + \frac{1}{2} \text{tr} \left\{ \Sigma_t(x, c) D_x^2 V_t(x) \right\} \right\} + \partial_t V_t(x) \end{aligned}$$

HJBE: Finite Horizon

Proposition (Necessity of the HJBE)

Define the (current) value function at time t by:

$$V_t(x) := \sup_{c \in \mathcal{C}(t,x)} \mathbb{E}_{t,x} \left[\int_t^T e^{-\rho(s-t)} u(c_s) ds + U(X_T^x) \right].$$

Suppose $V_t(x)$ exists and is $C^{1,2}$. Then $V_t(x)$ satisfies the HJBE:

$$\begin{aligned} \rho V_t(x) &= \max_{c \in \mathcal{C}(x)} \left\{ u(c) + \sum_{j=1}^n \mu_{t,j}(x, c) \partial_{x_j} V_t(x) + \frac{1}{2} \sum_{i,j=1}^n \Sigma_{t,jk}^2(x, c) \partial_{x_j x_k} V_t(x) \right\} + \partial_t V_t(x) \\ &= \sup_{c \in \mathcal{C}(x)} \left\{ u(c) + \mu_t(x, c) \cdot D_x V_t(x) + \frac{1}{2} \text{tr} \{ \Sigma_t(x, c) D_x^2 V_t(x) \} \right\} + \partial_t V_t(x) \\ V_T(x) &= U(x) \end{aligned}$$

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Multivariate Poisson Processes

- ▶ Now, suppose that $X_t \in \mathbb{R}^n$ and follows the Itô process:

$$dX_t = \mu_t(X_t, c_t)dt + h_t(X_t, c_t)dN_t$$

where

- ▶ N_t is an m -dimensional Poisson motion, for which element k has rate $\lambda_k(x)$,
- ▶ $\mu_t : \mathbb{R}^n \times C \rightarrow \mathbb{R}^n$ is the drift, and
- ▶ $h_t : \mathbb{R}^n \times C \rightarrow \mathbb{R}^{n \times m}$ is the jump size.

HJBE: Infinite Horizon

Proposition (Necessity of the HJBE)

Define the (current) value function at time t by:

$$V_t(x) := \sup_{c \in \mathcal{C}(t,x)} \mathbb{E}_{t,x} \left[\int_t^\infty e^{-\rho(s-t)} u(c_s) ds \right].$$

Suppose $V_t(x)$ exists and is $C^{1,2}$. Then $V_t(x)$ satisfies the HJBE:

$$\begin{aligned} \rho V_t(x) &= \max_{c \in \mathcal{C}(x)} \left\{ u(c) + \sum_{j=1}^n \mu_{t,j}(x, c) \partial_{x_j} V_t(x) + \sum_{k=1}^m \lambda_k(x) (V_t(X_t + h_{t,k}(X_t)) - V_t(X_t)) \right\} + \partial_t V_t(x) \\ &= \sup_{c \in \mathcal{C}(x)} \left\{ u(c) + \mu_t(x, c) \cdot D_x V_t(x) + \sum_{k=1}^m \lambda_k(x) (V_t(X_t + h_{t,k}(X_t)) - V_t(X_t)) \right\} + \partial_t V_t(x) \end{aligned}$$

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Environment

- ▶ Continuous time, infinite horizon economy with one consumption good.
- ▶ Infinitely lived entrepreneur gets flow utility $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, where c is flow goods consumption.
- ▶ Can save into risk free bond. Bond holdings, b_t , generate flow output at interest rate r :

$$dy_t^b = r b_t dt$$

- ▶ Can also operate a risky, CRS technology:
 - ▶ Requires capital, denoted by k_t , for production
 - ▶ Entrepreneur can convert capital into goods and goods into capital at a 1-1 ratio.
(So, implicitly price of capital $p_k = 1$)
 - ▶ Capital stock, k_t , generates flow output by the stochastic process:

$$dy_t^k = \bar{R} k_t dt + \sigma k_t dW_t$$

Entrepreneur's Budget Constraint

- ▶ Denote entrepreneur wealth by $a_t := b_t + k_t$, where:
 - ▶ b_t is the entrepreneur's bond holdings
 - ▶ k_t is the entrepreneur's capital holdings
- ▶ Let $\theta_t := k_t/a_t$ denote the portfolio share in capital. Then, the evolution of wealth is:

$$\begin{aligned} da_t &= dy_t^b + dy_t^k - c_t dt \\ &= r b_t dt + \bar{R} k_t dt + \sigma k_t dW_t - c_t dt \\ &= r(a_t - k_t) dt + \bar{R} k_t dt + \sigma k_t dW_t - c_t dt \\ &= r a_t dt + (\bar{R} - r) k_t dt + \sigma k_t dW_t - c_t dt \\ &= \left(r + (\bar{R} - r) \frac{k_t}{a_t} \right) a_t dt - c_t dt + \sigma \frac{k_t}{a_t} a_t dW_t \\ &= ((r + (\bar{R} - r) \theta_t) a_t - c_t) dt + \sigma \theta_t a_t dW_t \end{aligned}$$

Entrepreneur Problem

- ▶ The entrepreneur's problem is to choose controls, $\mathbf{c} = (c_t)_{t \geq 0}$ and $\theta = (\theta_t)_{t \geq 0}$, to solve:

$$V(x) := \sup_{\mathbf{c}, \theta} \mathbb{E} \left[\int_0^\infty e^{-\rho s} u(c_s) ds \right]$$

subject to the dynamic constraint on the evolution of their wealth a_t :

$$\begin{aligned} da_t &= \left((r + (\bar{R} - r)\theta_t) a_t - c_t \right) dt + \sigma \theta_t a_t dW_t \\ a_0 &= a \end{aligned}$$

- ▶ Q. Is this problem stationary?
- ▶ Q. Will the solution be Markovian?

Hamilton-Jacobi-Bellman Equation

- ▶ Let $V(a)$ denote the value function.
- ▶ Then, $V(a)$ satisfies the HJBE:

$$\rho V(a) = \max_{c, \theta} \left\{ u(c) + V'(a)(r + (\bar{R} - r)\theta)a - c + \frac{1}{2}\sigma^2\theta^2 a^2 V''(a) \right\}$$

- ▶ FOCs:

$$\begin{aligned}(c) : \quad & u'(c) = V'(a) \\ (\theta) : \quad & \theta = \left(\frac{\bar{R} - r}{\sigma^2} \right) \left(-\frac{V'(a)}{aV''(a)} \right)\end{aligned}$$

- ▶ Q. How do we interpret the FOC for θ condition?

Solving the HJBE: Guess and Verify

- Utility function form:

$$u(c) = (c^{1-\gamma})/(1-\gamma), \quad u'(c) = c^{-\gamma}$$

- Guess value function of the form (for unknown constant ψ):

$$V(a) = \psi^{-\gamma} \left(\frac{a^{1-\gamma}}{1-\gamma} \right) \Rightarrow V'(a) = \psi^{-\gamma} a^{-\gamma} \Rightarrow V''(a) = -\gamma \psi^{-\gamma} a^{-\gamma-1}$$

- FOC for c becomes:

$$c = \psi a$$

- FOC for θ becomes independent of a :

$$\theta = \frac{1}{\gamma} \left(\frac{\bar{R} - r}{\sigma^2} \right)$$

Solving the HJBE: Guess and Verify

- The HJBE becomes:

$$\begin{aligned}\rho V(a) &= \max_{c, \theta} \left\{ u(c) + V'(a)(r + (\bar{R} - r)\theta)a - c + \frac{1}{2}\sigma^2\theta^2a^2V''(a) \right\} \\ \Rightarrow \rho \frac{\psi^{-\gamma}a^{1-\gamma}}{1-\gamma} &= \frac{(\psi a)^{1-\gamma}}{1-\gamma} + \psi^{-\gamma}a^{-\gamma}(r + (\bar{R} - r)\theta)a - \psi a \\ &\quad - \frac{1}{2}\sigma^2\theta^2a^2\gamma\psi^{-\gamma}a^{-\gamma-1} \\ \Rightarrow \psi &= \frac{1}{\gamma} \left[\rho - (1-\gamma) \left(r + \frac{1}{2\gamma} \left(\frac{\bar{R} - r}{\sigma} \right)^2 \right) \right]\end{aligned}$$

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Application: Portfolio Problem

Application: The Consumption Saving Problem

- ...with Poisson Shocks

- ...with a Borrowing Constraint

Numerical Techniques

- Finite Difference

- Deep Learning

Household Consumption-Saving Problem

- A household has wealth, a , and labour, y . They choose consumption $\mathbf{c} = (c_t)_{t \geq 0}$ to solve:

$$\begin{aligned} V(a_0, y_0) &\equiv \max_{\mathbf{c}} \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} u(c_t) dt \right] \quad s.t. \\ a_t &= s(a_t, y_t, c_t; r_t, w_t) dt \\ &\equiv (w_t y_t + r_t a_t - c_t) dt \\ y_t &\in \{y_1, y_2\}, \text{ follows Poisson Process} \\ c_t &\geq 0 \end{aligned}$$

- c_t : consumption
- u : utility function, $u' > 0$, $u'' < 0$ (E.g. $u(c) = c^{1-\gamma}/(1-\gamma)$)
- ρ : discount rate
- r_t : interest rate, w_t : wage rate
- y_t : hours worked ($y_1 < y_2$); switching intensities $\{\lambda_1, \lambda_2\}$
- a_t : assets (claims on physical capital)

Hamilton-Jacobi-Bellman Equation (HJBE)

State vector: $\mathbf{x} \equiv \{a, y\}$; Value function: $V_t(a, y)$ on transition path and $V(a, y)$ at steady state.

- The HJBE equation for the steady state $V(a, y)$ is:

$$\rho V(a, y) = \max_{c \geq 0} \left\{ u(c) + \partial_a V(a, y)(wy + ra - c) + \lambda(y)(V(a, \check{y}) - V(a, y)) \right\}$$

where $\check{y}$ is complement of y and the first order condition is $u'(c) - \partial_a V(a, y) = 0$.

- The HJBE for $t \in [0, T)$ along the transition path to the steady state $V(a, y)$ is:

$$t \leq 0 : \rho V_t(a, y) = \max_{c \geq 0} \left\{ u(c) + \partial_a V_t(a, y)(w_t y + r_t a - c) + \lambda(y)(V_t(a, \check{y}) - V_t(a, y)) + \partial_t V_t(a, y) \right\}$$

$$t = T : V_T(a_T, y_T) = V(a_T, y_T)$$

where $\check{y}$ is complement of y and the first order condition is $u'(c) - \partial_a V_t(a, y) = 0$.

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- c_t : consumption
- u : utility function, $u' > 0$, $u'' < 0$ (E.g. $u(c) = c^{1-\gamma}/(1-\gamma)$)
- ρ : discount rate
- r_t : interest rate, w_t : wage rate
- y_t : hours worked ($y_1 < y_2$); switching intensities $\{\lambda_1, \lambda_2\}$
- a_t : assets (claims on physical capital)
- $\underline{a} > -\infty$: borrowing limit (E.g. $\underline{a} = 0$)

Hamilton-Jacobi-Bellman Equation

- ▶ Technically, this problem does not have a “classical solution” because the value function is not differentiable when the borrowing constraint binds
- ▶ Instead, need to show that a “viscosity” solution exists. This is beyond the scope of this course (and I recommend smooth functions), but you can look at the suggested references.
- ▶ For the viscosity solution, the steady-state HJBE equation is:

$$\rho V(a, y) = \max_{c \geq 0} \left\{ u(c) + \partial_a V(a, y)(wy + ra - c) + \lambda(y)(V(a, y') - V(a, y)) \right\}$$
$$\text{s.t. } u'(wy + r\underline{a}) \leq \partial_a V_t(\underline{a}, y)$$

- ▶ NOTE: The borrowing constraint appears as a boundary condition. Why?
Agents are still optimizing on the boundary but the borrowing constraint cannot be violated.

$$u'(c^*) = \partial_a V_t(a, y), \quad \text{for } a > \underline{a}$$
$$u'(wy + r\underline{a}) \leq \partial_a V_t(\underline{a}, y), \quad \text{for } a = \underline{a}, \quad \text{since } \partial_a V_t(\underline{a}, y) = u'(c^*(\underline{a})) \text{ and } c^*(\underline{a}) \leq wy + r\underline{a}$$

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Example 2: Hamilton-Jacobi-Bellman Equation

(From the Consumption-Saving Problem)

- **Model:** Suppose $V(a, y)$ satisfies the (stationary) HJBE for $a \geq \underline{a}$:

$$0 = -\rho V(a, y) + \max_{c \geq 0} \left\{ u(c) + \partial_a V(a, y)(wy + ra - c) + \lambda(y)(V(a, \tilde{y}) - V(a, y)) \right\}$$
$$s.t. \quad u'(wy + r\underline{a}) \leq \partial_a V(\underline{a}, y)$$

which has the first order condition for $a > \underline{a}$:

$$u'(c) = \partial_a V(a, y)$$

- **Task:** Solve for $V(a, y)$.
- Note: this problem is non-linear because c is a non-linear function of $\partial_a V(a, y)$

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Discretisation

- ▶ Discretize the a -dimension into I equispaced grid points with distance Δa between the grid points: $\{a_i : i = 1, \dots, I\}$
- ▶ The y -dimension is already discretised: $\{y_j : j = 1, 2\}$
- ▶ Denote $v_{i,j} \equiv v(a_i, y_j)$ and $v'_{i,j} \equiv \partial_a v(a_i, y_j)$
- ▶ Forward and backward differences:

$$v'_{i,j} \approx \frac{v_{i+1,j} - v_{i,j}}{\Delta a} \equiv v'_{i,j,F} \quad (4)$$

$$v'_{i,j} \approx \frac{v_{i-1,j} - v_{i,j}}{\Delta a} \equiv v'_{i,j,B} \quad (5)$$

HJBE: Finite Difference Approximation

- ▶ The finite difference approximation to the stationary HJBE is:

$$\begin{aligned} 0 &= -\rho v_{i,j} + u(c_{i,j}) + v'_{i,j}(wy_j + ra_i - c_{i,j}) + \lambda_j(v_{i,-j} - v_{i,j}), \quad 1 \leq i \leq I \quad j \in \{1, 2\}, \\ c_{i,j} &= (u')^{-1}(v'_{i,j}), \quad 1 \leq i \leq I \quad j \in \{1, 2\} \end{aligned}$$

- ▶ Issues:

- ▶ HJBE is non-linear because optimal choice of $c_{i,j}$ depends upon $v_{i,j}$.
- ▶ This means we need to undertake “value function iteration”.
- ▶ Need to decide whether to use forward or backward difference for $v'_{i,j}$

How can we handle non-linearity?

Value Function Iteration Using a “False” Time Derivative

- ▶ We work with the non-stationary version of the HJBE (rearranged from Lecture 3):

$$-\partial_t V_t(a, y) = -\rho V_t(a, y) + \max_{c \geq 0} \left\{ u(c) + \partial_a V_t(a, y)(wy + ra - c) + \lambda(y)(V_t(a, \tilde{y}) - V_t(a, y)) \right\}$$
$$s.t. \quad u'(wy + r\underline{a}) \leq \partial_a V(\underline{a}, y)$$

- ▶ Idea: start from arbitrary $V_T(a, y)$ then iterate back in time until V_t converges to V .
- ▶ Discretization: in the space and time dimensions is (for $1 \leq i \leq I, j \in \{1, 2\}, t \leq T$):

$$-\frac{v_{i,j,t} - v_{i,j,t-1}}{\Delta} = -\rho v_{i,j,*} + u(c_{i,j,*}) + v'_{i,j,*}(wy_j + ra_i - c_{i,j,*}) + \lambda_j(v_{i,-j,*} - v_{i,j,*}),$$
$$c_{i,j,*} = (u')^{-1}(v'_{i,j,*}),$$

where Δ is the time step and $*$ could be t or $t - 1$ on the RHS.

- ▶ Notation change: we denote the value function at iteration step n by $v_{i,j}^n := v_{i,j,T-n}$.

HJBE Iteration: Forward Euler Method (Explicit; t on RHS)

- ▶ Start with an initial guess $\{v_{i,j}^0\}_{1 \leq i \leq I, j \in \{1,2\}}$
- ▶ At iteration $n = 1, 2, \dots$ update guess $\{v_{i,j}^n\}_{1 \leq i \leq I, j \in \{1,2\}}$ by **Forward Euler Method**:

$$\frac{v_{i,j}^{n+1} - v_{i,j}^n}{\Delta} = u(c_{i,j}^n) + (v_{i,j}^n)'(wy_j + ra_i - c_{i,j}^n) + \lambda_j(v_{i,-j}^n - v_{i,j}^n) - \rho v_{i,j}^n$$
$$c_{i,j}^n = (u')^{-1}((v_{i,j}^n)'), \quad 1 \leq i \leq I, \quad j \in \{1, 2\}$$

- ▶ Interpretation: Correcting for guess error (or iterating backwards):

$$v_{i,j}^{n+1} = v_{i,j}^n + \underbrace{\Delta(u(c_{i,j}^n) + (v_{i,j}^n)'(wy_j + ra_i - c_{i,j}^n) + \lambda_j(v_{i,-j}^n - v_{i,j}^n) - \rho v_{i,j}^n)}_{\text{Error in the stationary HJBE for current guess } v^n}$$

- ▶ This is an **explicit scheme** because we have an explicit solution for v^{n+1}
- ▶ Easier to code but less stable

HJBE: Mixed Euler Method (Semi-Implicit; t and $t - 1$ on RHS)

- ▶ Start with an initial guess $\{v_{i,j}^0\}_{1 \leq i \leq I, j \in \{1,2\}}$
- ▶ At iteration $n = 1, 2, \dots$, update guess $\{v_{i,j}^n\}_{1 \leq i \leq I, j \in \{1,2\}}$ by **Mixed Euler Method**:

$$\frac{v_{i,j}^{n+1} - v_{i,j}^n}{\Delta} = u(c_{i,j}^n) + (v_{i,j}^{n+1})'(wy_j + ra_i - c_{i,j}^n) + \lambda_j(v_{i,-j}^{n+1} - v_{i,j}^{n+1}) - \rho v_{i,j}^{n+1}$$
$$c_{i,j}^n = (u')^{-1}((v_{i,j}^n)'), \quad 1 \leq i \leq I, \quad j \in \{1, 2\}$$

- ▶ This is an explicit scheme in $c_{i,j}^n$ but is implicit in other variables.
- ▶ This scheme is suggested by [Achdou et. al. \(2018\)](#).
(And used in my sample code.)

HJBE: Choice of Forward or Backwards Derivative

- ▶ Candler (1999) & Achdou et. al. (2019) suggest an **upwind scheme**:
 - ▶ Use the forward difference when the drift of the state is positive
 - ▶ Use the backward difference when the drift of the state is negative
 - ▶ Intuition: Minimises the error when discontinuities are propagated across the state space
- ▶ Upwind scheme leads to the following finite difference equation:

$$\frac{v_{i,j}^{n+1} - v_{i,j}^n}{\Delta} = u(c_{i,j}^n) + (v_{i,j,F}^{n+1})'[s_{i,j}^n]^+ + (v_{i,j,B}^{n+1})'[s_{i,j}^n]^- + \lambda_j(v_{i,-j}^{n+1} - v_{i,j}^{n+1}) - \rho v_{i,j}^{n+1}$$

- ▶ $x^+ = \max\{x, 0\}$, $x^- = -\min\{x, 0\}$
- ▶ $s_{i,j}^n = wy_j + ra_i - c_{i,j}^n$
- ▶ $c_{i,j}^n = (u')^{-1}[(v_{i,j}^n)']$

HJBE: Matrix Form

- ▶ Like in the finite difference lecture, we can stack the system in matrix form.
- ▶ We can then use linear algebra to solve the differential equation.

HJBE: Matrix Form: Boundary Conditions

- ▶ Upwind scheme adds “shadow” nodes at $i = 0$ & $i = J + 1$ aren't solved for.
- ▶ Lower boundary:
 - ▶ If $s_{1,j,F} > 0$, then upwind scheme chooses $v'_{1,j,F}$, which does not need the value v_0 at the shadow node $i = 0$.
 - ▶ If $s_{1,j,F} \leq 0$, then set the backward derivative to enforce non-negative saving constraint:

$$v'_{1,j,B} = u'(wy_j + ra_1) \Leftrightarrow s_{1,j,B} = 0, \quad j = 1, 2$$

- ▶ Upper boundary:
 - ▶ If $a_J > a_{max}$, then $s_{I,j,F} \leq 0$ (since agents stop saving once $a \geq a_{max}$) and so the backward scheme $v'_{1,j,B}$ is chosen.
 - ▶ However, Achdou et. al. (2019) enforce this anyway by setting

$$u'_{I,j,F} = u'(wy_j + ra_I) \Leftrightarrow s_{I,j,F} = 0, \quad j = 1, 2$$

HJBE: Matrix Form

- Finite difference approximation consists of $2 \times J$ linear equations:

$$\begin{aligned} \frac{v_{i,j}^{n+1} - v_{i,j}^n}{\Delta} = & u(c_{i,j}^n) + \left(\frac{v_{i+1,j}^{n+1} - v_{i,j}^{n+1}}{\Delta a} \right) [s_{i,j,F}^n]^+ + \left(\frac{v_{i,j}^{n+1} - v_{i-1,j}^{n+1}}{\Delta a} \right) [s_{i,j,B}^n]^- \\ & + \lambda_j (v_{i,-j}^{n+1} - v_{i,j}^{n+1}) - \rho v_{i,j}^{n+1}, \quad i \in \{1, 2\}, 1 \leq j \leq I \end{aligned}$$

- Collecting terms on the RHS gives:

$$\frac{v_{i,j}^{n+1} - v_{i,j}^n}{\Delta} + \rho v_{i,j}^{n+1} = u(c_{i,j}^n) + v_{i-1,j}^{n+1} \alpha_{i,j} + v_{i,j}^{n+1} \beta_{i,j} + v_{i+1,j}^{n+1} \gamma_{i,j} + v_{i,-j}^{n+1} \lambda_j$$

where:

$$\begin{aligned} \alpha_{i,j} &= -\frac{[s_{i,j,B}^n]^-}{\Delta a}, \\ \beta_{i,j} &= -\frac{[s_{i,j,F}^n]^+}{\Delta a} + \frac{[s_{i,j,B}^n]^-}{\Delta a} + \lambda_j, \\ \gamma_{i,j} &= \frac{[s_{i,j,F}^n]^+}{\Delta a} \end{aligned}$$

HJBE: Matrix Form

- Observe that:

$$\alpha_{1,j} = -\frac{(s_{1,j,B})^-}{\Delta a} = 0, \quad \beta_{I,j} = \frac{(s_{I,j,B})^+}{\Delta a} = 0$$

so $v_{0,j}^{n+1}$ and $v_{I+1,j}^{n+1}$ are never used.

- The system consists of $2 \times I$ equations. We stack the variable in the y dimension. That is, let

$$v^n = \begin{bmatrix} v_{1,1}^n \\ v_{2,1}^n \\ \vdots \\ v_{I,1}^n \\ v_{1,2}^n \\ v_{2,2}^n \\ \vdots \\ v_{I,2}^n \end{bmatrix}, \quad u^n = \begin{bmatrix} u(c_{1,1}^n) \\ u(c_{2,1}^n) \\ \vdots \\ u(c_{I,1}^n) \\ u(c_{1,2}^n) \\ u(c_{2,2}^n) \\ \vdots \\ u(c_{I,2}^n) \end{bmatrix},$$

HJBE: Matrix Form

- Then, the updating equation becomes:

$$\frac{1}{\Delta}(v^{n+1} - v^n) + \rho v^{n+1} = u^n + A^n v^{n+1}$$

where:

$$A^n = \begin{bmatrix} \beta_{1,1} & \gamma_{1,1} & 0 & \dots & 0 & \lambda_1 & 0 & \dots & 0 & 0 & 0 \\ \alpha_{2,1} & \beta_{1,1} & \gamma_{2,1} & 0 & \dots & 0 & \lambda_1 & 0 & \dots & 0 & 0 \\ 0 & \alpha_{2,1} & \beta_{1,1} & \gamma_{2,1} & 0 & \dots & 0 & \lambda_1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \ddots & 0 & \beta_{I,1} & \gamma_{I,1} & 0 & 0 & 0 & \dots & 0 & \lambda_1 \\ \lambda_2 & 0 & \dots & 0 & 0 & \beta_{1,2} & \gamma_{1,2} & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 & \alpha_{2,2} & \beta_{2,2} & \gamma_{2,2} & 0 & \dots & 0 \\ 0 & 0 & \lambda_2 & 0 & \dots & 0 & \alpha_{2,2} & \beta_{2,2} & \gamma_{2,2} & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \lambda_2 & 0 & \dots & 0 & 0 & \alpha_{I,2} & \beta_{I,2} \end{bmatrix}$$

HJBE: Matrix Form

- This system can in turn be written as:

$$\mathbf{B}^n v^{n+1} = b^n$$

where:

$$\mathbf{B}^n = \left(\frac{1}{\Delta} + \rho \right) \mathbf{I} - \mathbf{A}^n,$$
$$b^n = u^n + \frac{1}{\Delta} v^n$$

Summary: HJBE Algorithm

1. Guess $v_{i,j}^0$, $i = 1, \dots, I$, $j = 1, 2$
2. For iterations $n = 1, 2, \dots$ follow:
 - 2.1 Compute $(v_{i,j}^n)'$ using
 - 2.2 Compute c^n from $c_{i,j}^n = (u')^{-1}[(v_{i,j}^n)']$
 - 2.3 Find v^{n+1} by solving:

$$\mathbf{B}^n v^{n+1} = b^n \quad \Rightarrow \quad v^{n+1} = (\mathbf{B}^n)^{-1} b^n$$

- 2.4 If v^{n+1} is close enough to v^n , then stop. Else, return to step 1.

Sample Code

- ▶ `Julia-Aiyagari-FiniteDiff` folder.
- ▶ I am going to ask you to extend this in the next homework.

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A Different Spirit

- ▶ The finite difference approach tries to reduce the non-linear PDE to linear steps.
- ▶ Deep learning machinery attempts to solve the non-linear PDE directly.

Softening the Borrowing Constraint

- ▶ You can try the hard borrowing constraint but you can also soften the constraint.
- ▶ To make the problem more tractable, we introduce a penalty function ψ at the left boundary, replacing the agent flow utility by:

$$U(a_t, c_t) = u(c_t) + \mathbf{1}_{a_t \leq \underline{a}} \psi(a_t) \quad (6)$$

- ▶ E.g. the quadratic penalty: $\psi(a) = -\frac{1}{2}\kappa(a - \underline{a})^2$ where κ is a positive constant.

The HJBE Equation

- Each household chooses c to solve the HJBE:

$$0 = \mathcal{L}^V(a, n) = \max_c \left\{ -\rho V(a, n) + u(c) + \mathbf{1}_{a \leq \underline{a}} \psi(a) \right. \\ \left. + \partial_a V(a, n) s(c, a, n, r, w) + \lambda(n)(V(a, \check{n}) - V(a, n)) \right\}$$

and the optimal control, c^* , is characterised by the first order condition:

$$\partial_a V(a, n) = u'(c^*(a, n)).$$

The Euler Equation

- ▶ It is often easier to approximate the derivative of the value function. Why?
- ▶ Define $W(a, n) := \partial_a V(a, n)$.
- ▶ Applying the envelope theorem to the HJBE and imposing the FOC, we get:

$$\begin{aligned} 0 = \mathcal{L}^W(a, n) = & (r - \rho)W(a, n) + \mathbf{1}_{a \leq \underline{a}}\psi(a) \\ & + \partial_a W(a, n)s(c, a, n, r, w) + \lambda(n)(W(a, \check{n}) - W(a, n)) \} \end{aligned}$$

EMINN Algorithm

1. Approximate W by NN: $W(x) \approx \hat{W}(x; \theta)$, where θ are NN parameters and $x = (a, n)$.
2. Make initial parameter guess θ^0 .
3. At iteration n with guess θ^n :
 - 3.1 Generate M sample points, $S = \{(x_m)\}_{m \leq M}$ for evaluating the master equation error.
 - 3.2 Calculate the average error in the master equation for the sample:

$$\mathcal{E}(\theta^n, S) := \frac{1}{M} \sum_{m \leq M} |\mathcal{L}^W(x_m)|^2$$

- 3.3 Update the the parameters using “deep learning” toolkit. For example:

$$\theta^{n+1} = \theta^n - \alpha_n D_\theta \mathcal{E}(\theta^n, S^n)$$

- 3.4 Repeat until training loss $\mathcal{E}(\theta^n, S^n) \leq \epsilon$ (or test on a validation set).

Summary and Lessons

- ▶ Characterized continuous time optimal control problem as HJBE.
- ▶ Lesson is that continuous time optimal control involves solving differential equations.
- ▶ We studied two solution techniques: finite difference and neural networks.

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